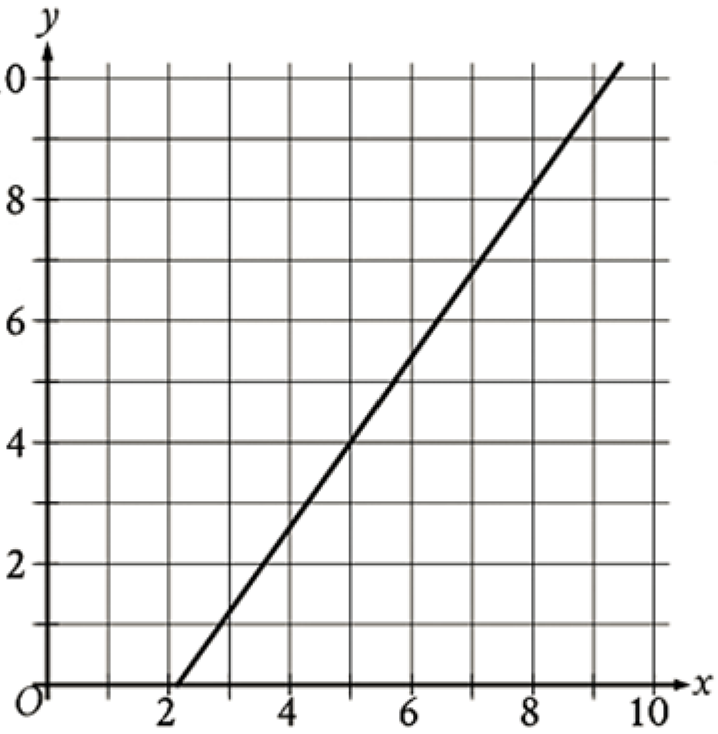
Math Module 2

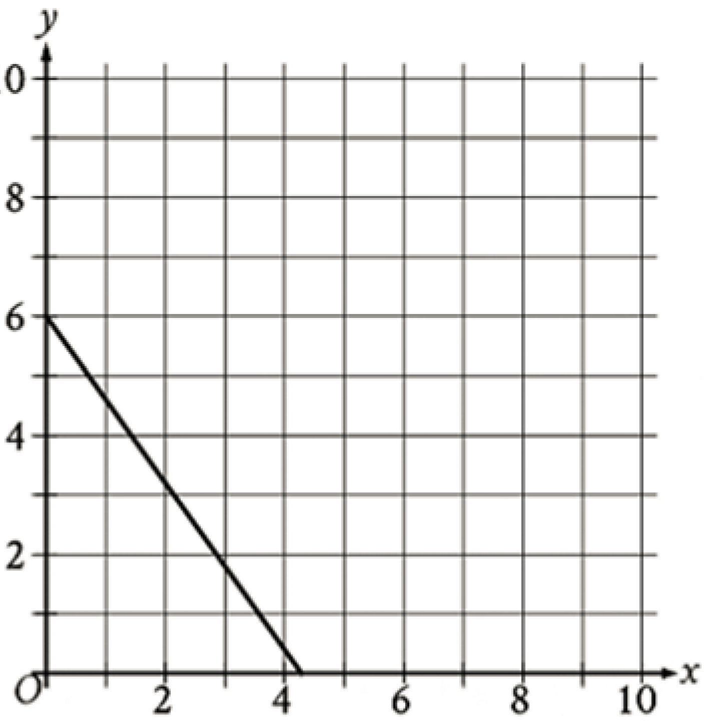
1

Which of the following is a graph of a decreasing exponential function?

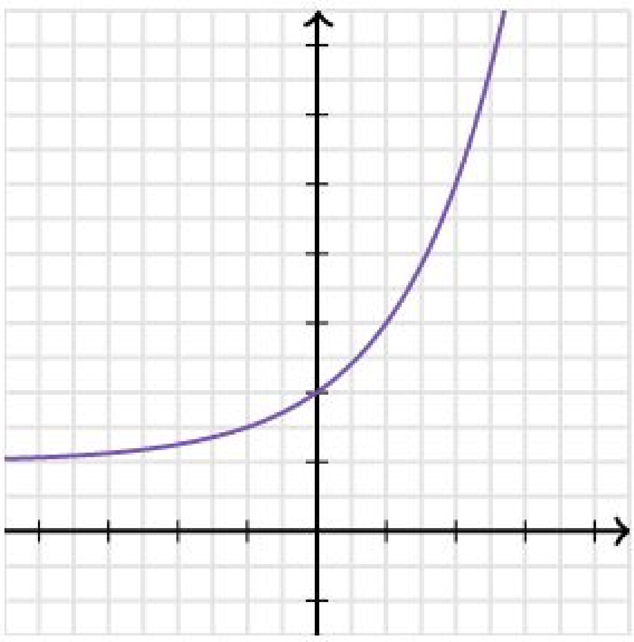
A)

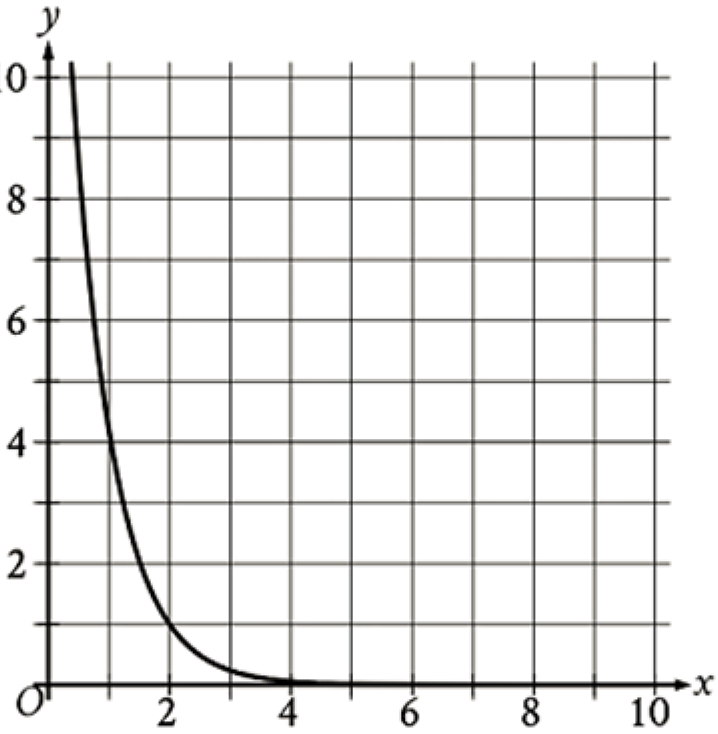


B)



C)



D) 

2

What is the y-intercept of the graph of $y= 6x^2 + 3x + 4$ in the xy-plane?

A) (0,3)

B) (0,4)

C) (0,6)

D) (0,7)

3

If $2x+5=9$, what is the value of $8x-4$?

4

The product of a positive number x and the number that is 1 less than x is equal to 12. What is the value of x?

A) 0.5

B) 4

C) 11

D) 13

5

A cylinder has a diameter of 6 inches and a height of 24 inches. What is the volume, in cubic inches, of the cylinder?

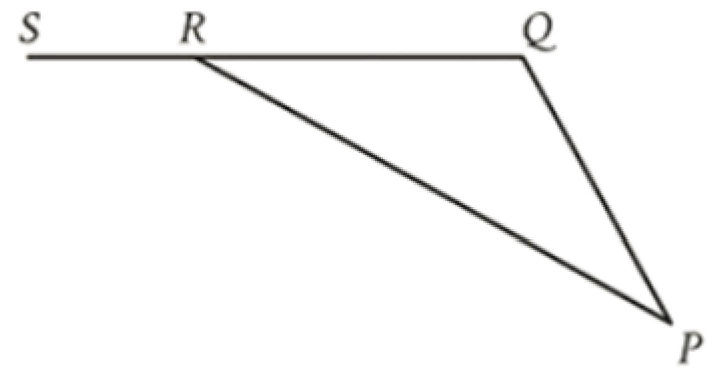
A) 9π

B) 144π

C) 216π

D) 864π

6.



Note: Figure not drawn to scale.

In triangle PQR, QR is extended to point S. The measure of ∠PQR is 114°, and the measure of ∠PRS is 158°. What is the measure of ∠QPR?

A) 66°

B) 44°

C) 33°

D) 22°

7

Sarah used wax to make candles. The function f(x) = - 12x+ 60 approximates the weight, in ounces, of wax Sarah had remaining after making a candles. Which statement is the best interpretation of the y-intercept of the graph of y= f(x) in the xy-plane in this context?

A) Sarah used approximately 12 ounces of wax for each candle.

B) Sarah had approximately 12 ounces of wax when she began to make the candles.

C) Sarah had approximately 60 ounces of wax when she began to make the candles.

D) Sarah used approximately 60 ounces of wax for each candle.

8

$$ak = \frac{b}{11}(14 + m)$$

The given equation relates the positive variables a, b, k, and m. Which equation correctly expresses m in terms of a, b, and k?

A) $\frac{11(ak-14)}{b}$

B) $ak - \frac{14-b}{11}$

C) $\frac{ak-14b}{11}$

D) $\frac{11ak}{b} - 14$

9

The function g is defined by

g(x) = 10^{3x-7}

What is the value of a when g(a) is equal to 10,000?

A) 3

B) 10/3

C) 11/3

D) 4

10 If $\frac{x}{y} = 2$ and $\frac{5x}{ty} = 50$, what is the value of t?

A) $\frac{1}{5}$

B) $\frac{2}{5}$

C) $\frac{5}{2}$

D) 5

11

A piece of string with a length of 30 inches is cut into two parts. One part has a length of x inches, and the other part has a length of y inches. The value of x is 3 more than 2 times the value of y. What is the value of x?

12

Which equation has no solution?

A) $-15x - \frac{1}{6} = -15x - \frac{1}{6}$

B) $-15x - \frac{1}{6} = -15x + \frac{1}{6}$

C) $-15x - \frac{1}{6} = 15x - \frac{1}{6}$

D) $-15x - \frac{1}{6} = 15x + \frac{1}{6}$

13

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 9.40 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

A) 16.92

B) 48.92

C) 474.75

D) 506.75

14

$y = -0.5$

$y = x^2 + 10x + a$

In the given system of equations, a is a positive integer constant. The system has no real solutions. What is the least possible value of a?

15

An environmental scientist is investigating the volatility of 175 organic compounds by finding the boiling point of each compound. The mean boiling point of all 175 compounds is 183 degrees Celsius. The scientist classifies each of these compounds as either semivolatile or volatile. Of the 175 compounds, 50 compounds were classified as semi-volatile, and these 50 compounds have a mean boiling point of 338 degrees Celsius. The remaining125 compounds were classified as volatile. What is the mean boiling point, in degrees Celsius, of the 125 compounds classified as volatile compounds?

16. $f(x) = 28(1.30)^{x/4}$

For the given function f, the value of f(x) increases by p% for every increase of x by 8. What is the value of p?

A) 30

B) 41

C) 60

D) 69

17

A carpenter charges a flat rate of $306 for the first 3 hours of work and $85 for each additional hour of work. Which equation gives the total amount y, in dollars, that the carpenter charges for x hours of work, where x >3?

A) y = 85x + 51

B) y = 306x + 85

C) y = 306x + 561

D) y = 85x + 306

18

$$\sqrt[3]{285n}\left(\sqrt[4]{285n}\right)^2$$

For what value of x is the given expression equivalent to $(285n)^{12x}$, where $n > 1$?

19

	0–9 years	10–19 years	20+ years	Total
Group A	15	18	7	40
Group B	6	7	27	40
Group C	19	15	6	40
Total	40	40	40	120

The table summarizes the distribution of age and assigned group for 120 participants in a study. One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age?

A) 5/24

B) 5/16

C) 9/20

D) 5/8

20.

x	$g(x)$
-24	3
-9	0
16	5

The table shows three values of x and their corresponding values of g(x), where $g(x) = \frac{f(x)}{x+4}$ and f is a linear function. What is the y-intercept of the graph of y = f(x) in the xy-plane?

A) (0,36)

B) (0,9)

C) (0,4)

D) (0,-9)

21

The mass of object A is 481% of the mass of object B, and the mass of object A is 0.074% of the mass of object C. If the mass of object C is p% of the mass of object B, what is the value of $\frac{p}{1000}$?

22

$$\frac{1}{cx} = \frac{x}{128} + \frac{1}{c}$$

In the given equation, c is a constant. If the equation has exactly one solution, what is the value of c?